A FREE SHEET · DR SUDEEP ROPLEKAR, RADIOLOGIST

Your first fitness number

Six minutes of walking, or twelve of running, on flat ground. Then one line of arithmetic and a table.

Verve

1 · PICK YOUR TEST

Walk six minutes as far as you can. If you already run, run twelve minutes. Flat ground: a track, a park loop, a quiet road.

2 · MEASURE THE DISTANCE

In metres. A 400 m track makes it easy; otherwise a phone map or a watch. Write it down the moment you stop.

3 · FIND YOUR TIER

Turn the distance into METs with the line below, then read your age row. One MET is you at rest; ten is ten times that.

> A heart condition, chest pain on exertion, or told not to exert yourself? Walk, do not run, and ask your doctor first.

If you ran twelve minutes

`VO2max = (metres - 504.9) ÷ 44.73`
`METs = VO2max ÷ 3.5`

2,400 m: (2400 − 504.9) ÷ 44.73 = 42.4, and 42.4 ÷ 3.5 = **12.1 METs**.

Cooper KH, JAMA, 1968.

If you walked six minutes

`VO2max = 61.1 - 11.1 × sex - 0.4 × age - 0.2 × weight kg + 0.02 × metres`
`sex: 1 for a man, 2 for a woman · METs = VO2max ÷ 3.5`

A man of 45, 80 kg, 600 m: 61.1 − 11.1 − 18 − 16 + 12 = 28.0, and 28.0 ÷ 3.5 = **8.0 METs**.

Hong et al., 2019, young adults. Over 60 the app uses a line that also takes height (Šagát et al., 2023).

Men, METs by age

Age	Low	Below avg	Above avg	High	Elite
20–29	to 9.3	9.4–10.3	10.4–12.0	12.1–13.2	13.3+
30–39	to 8.9	9.0–10.0	10.1–11.6	11.7–12.8	12.9+
40–49	to 7.9	8.0–9.9	10.0–11.9	12.0–13.9	14.0+
50–59	to 6.9	7.0–8.9	9.0–10.9	11.0–12.9	13.0+
60–69	to 5.9	6.0–7.9	8.0–9.9	10.0–11.9	12.0+
70+	to 4.9	5.0–6.9	7.0–8.9	9.0–10.9	11.0+

Women, METs by age

Age	Low	Below avg	Above avg	High	Elite
20–29	to 6.6	6.7–8.2	8.3–9.3	9.4–10.5	10.6+
30–39	to 6.4	6.5–7.6	7.7–8.9	9.0–10.1	10.2+
40–49	to 5.9	6.0–7.9	8.0–9.9	10.0–11.9	12.0+
50–59	to 4.9	5.0–6.9	7.0–8.9	9.0–10.9	11.0+
60–69	to 3.9	4.0–5.9	6.0–7.9	8.0–9.9	10.0+
70+	to 3.4	3.5–4.9	5.0–6.9	7.0–8.9	9.0+

Ages 40 and over: Mandsager K et al., JAMA Network Open, 2018, 122,007 patients at the Cleveland Clinic, treadmill tested. Under 40: Cooper Institute and ACSM norms, 1997, VO2max ÷ 3.5. In the Cleveland Clinic study, below average rather than above carried about the same risk of dying as smoking or diabetes. A tier is a place in a table, and it moves.

Your result. Keep this sheet, and test again.

Date	Walk or run	Metres	VO2max	METs	Tier

The number only means something next to the one before it. Test again after about eight weeks of training. Sooner is hard to tell from a good or bad day (Hickson et al., 1981). That is the window the app uses.

Written by Dr Sudeep Roplekar, radiologist, who built Verve to see his own numbers move against the thresholds the studies use. The app does this arithmetic for you and reads twenty other numbers the same way. The letters: roplekarsudeep.substack.com